# Taizoon Aliasgar Chunawala

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## Education

**Virginia Tech**, Blacksburg, VA — Aug 2021 – Expected May 2027
Ph.D., Mechanical Engineering
*Dissertation:* Structurally Augmented Optimal and Safety-Critical Control for Robotic Systems Under Dynamic and Environmental Uncertainties

**University of Southern California**, Los Angeles, CA — Dec 2019
M.S., Aerospace & Mechanical Engineering (Dynamics & Control)

**BITS Pilani**, Pilani, India — Jun 2016
B.E. (Hons.), Mechanical Engineering

## Technical Skills

**Control & Planning:** Nonlinear/Robust/Distributed MPC, Trajectory Optimization (Direct Collocation), Contact Planning, Whole-Body Control (QP), Optimal Control
**Estimation:** Kalman Filtering, Sensor Fusion (Kinematics + IMU, Vision + IMU), State Estimation
**Programming:** C++ (real-time, multithreaded), Python, MATLAB/Simulink
**Solvers & Libraries:** CasADi, IPOPT, OSQP, qpOASES, Pinocchio, Eigen
**Frameworks & Tools:** ROS/ROS2, MuJoCo, Raisim, OpenCV, Linux, Git, CMake
**Hardware:** Unitree Go2, Unitree A1, Kinova arm
**Modeling & Design:** Rigid-Body Dynamic Modeling, URDF, SolidWorks

## Research Experience

**Graduate Research Assistant** — Feb 2022 – Present
Agile Robots for Control, Optimization & Learning Lab, Virginia Tech, Blacksburg, VA

- **Multi-Rate NMPC for Wall-Supported Bipedal Locomotion of a Quadruped** *(lead author)*
  - Developed a layered multi-rate NMPC that **jointly plans footstep locations and single-rigid-body (SRB) trajectories**, achieving **2.9× higher success** than a Raibert-heuristic baseline across **250 randomized rough-terrain trials** in simulation (Unitree A1).
  - Solved the NMPC in **8.3 ms at 100 Hz** (CasADi/IPOPT), tracked by a **1 kHz** QP-based whole-body controller enforcing full-order dynamics.
  - Built a multithreaded real-time C++ stack and 1 kHz kinematics–IMU state estimator for **preliminary hardware deployment on a Unitree Go2**: crouch to wall-supported bipedal walking, balanced for 60+ s. [paper] [video]
- **Robust MPC for a Tremor-Suppression Exoskeleton** *(lead author)*
  - Designed a robust MPC that plans across the **worst-case tremor disturbances** of a convex uncertainty set built from tremor recorded in essential-tremor patients.
  - Introduced **state-dependent cost weighting** that prioritizes the disturbance driving tracking error, cutting RMSE **20–33%** vs. observer-based MPC and **up to 47%** on patient-derived tremor (simulation).
  - Reduced the 9-DoF arm–exoskeleton model to 2 wrist coordinates for real-time control (14.6 ms average solve per 20 ms step); stress-tested under sensor noise and 50% actuation latency. [video]
- **Decomposition-Based NMPC for Quadrupedal Loco-Manipulation** *(co-author)*
  - Implemented the real-time interface between a **60 Hz decomposition NMPC** planner and a **500 Hz QP-based WBC**, passing optimal ground reaction forces and interaction wrenches for tracking.
  - Supported hardware validation on a Unitree Go2 carrying a 4.4 kg Kinova arm under unmodeled terrain and 125% payload uncertainty. [video]

- **Safety-Critical Distributed NMPC for Multi-Robot Quadrupedal Teams** *(co-author)*
  - Co-developed a **100 Hz distributed nonlinear MPC** with safety constraints for Unitree A1 robot teams.
  - Diagnosed and fixed real-time data pipeline and one-step-delay communication issues across multi-threaded processes, enabling stable hardware experiments under latencies up to 6 ms. [video]
- **Adaptive MPC for Payload Transportation by Quadrupedal Robots** *(co-author)*
  - Designed payload-transportation experiments and ran additional hardware experiments on the Unitree A1 to address reviewer comments (payloads up to 109% of robot mass). [video]

**Research Assistant** Feb 2020 – Dec 2020

Dynamic Robotics & Control Lab, University of Southern California, Los Angeles, CA

- Formulated and solved **jumping trajectory optimization via direct collocation** for an 18-DOF quadruped on compliant terrain, achieving target jump heights and stance-to-flight transitions in simulation.

**Research Assistant** May 2018 – May 2019

Autonomous Microrobotic Systems Lab, University of Southern California, Los Angeles, CA

- Developed adaptive and predictive controllers for piezo-actuated flapping-wing microrobots, implemented in Simulink Real-Time for closed-loop flight experiments.

## Professional Experience

**Control Engineering Intern** Jun 2019 – Aug 2019

Polybee, Singapore

- Developed a vision-based localization system with **millimeter-level accuracy** for micro-quadrotors performing autonomous pollination in vertical farms.
- Fused vision-based localization with IMU data for robust state estimation and smooth trajectory tracking in confined indoor spaces.

**Operations Manager** Jul 2016 – Nov 2017

Tata Steel Global Wires, Tarapur, India

- Optimized production scheduling across wire grades and gauges, removing bottlenecks to improve line uptime and throughput.

## Selected Publications

**T. Chunawala** et al., "Multi-Rate Nonlinear Model Predictive Control for Wall-Supported Bipedal Locomotion of Quadrupedal Robots," *IEEE/RSJ IROS*, 2026. [arXiv]

**T. Chunawala** et al., "Robust Model Predictive Control of a Tremor-Suppression Exoskeleton under Tremor-Induced Disturbances," *ASME Journal of Dynamic Systems, Measurement, and Control*, accepted, 2026.

R. Sambhus, K. K. Mehta, A. M. Sadeghi, B. M. Imran, J. Kim, **T. Chunawala** et al., "A Nonlinear MPC Framework for Loco-Manipulation of Quadrupedal Robots With Non-Negligible Manipulator Dynamics," *IEEE Robotics and Automation Letters*, 2026.

B. M. Imran, J. Kim, **T. Chunawala** et al., "Safety-Critical and Distributed Nonlinear Predictive Controllers for Teams of Quadrupedal Robots," *IEEE Robotics and Automation Letters*, 2025.

L. Amanzadeh, **T. Chunawala** et al., "Predictive Control with Indirect Adaptive Laws for Payload Transportation by Quadrupedal Robots," *IEEE Robotics and Automation Letters*, 2024.

## Honors

**Pratt Fellowship**, Department of Mechanical Engineering, Virginia Tech 2021–22